

Categories, Proofs, and Processes: Assignment 3

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A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is

- **faithful** if each map $F_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{D}(F A, F B)$ is injective;
- **full** if each map $F_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{D}(F A, F B)$ is surjective

Question 1

For each function $f : C \rightarrow D$ in **Set**, S maps it to the functor

$$S f : \mathcal{C} \rightarrow \mathcal{D}$$

where $\mathcal{C}, \mathcal{D}$ are discrete categories in **Cat** with $|C|, |D|$ objects respectively.

$$\mathbf{Set} \xrightarrow{S} \mathbf{Cat}$$

$$\begin{array}{ccc} C & \xrightarrow{\quad} & C \\ f \downarrow & \xrightarrow{\quad} & \downarrow S f \\ D & \xrightarrow{\quad} & D \end{array}$$

Then $\forall a \in C, b \in D, (S a) \in \text{Ob}(\mathcal{C}), (S b) \in \text{Ob}(\mathcal{D})$, the mapping is

$$S f :: S a \mapsto S b \iff f :: a \mapsto b$$

$$S f :: 1_{(S a)} \mapsto 1_{(S b)} \iff f :: a \mapsto b$$

Proposition 1.1. $\forall f \in \text{Ar}(\mathbf{Set}), (S f) : \mathcal{C} \rightarrow \mathcal{D}$ satisfies functoriality.

Proof. Since there is a valid function where $\forall a \in C, f(a) = b$, $(S f)$ will map every object $(S a)$ in $\mathcal{C}$ to an $(S b)$ in $\mathcal{D}$.

There are no arrows between objects in the discrete categories, only identity arrows. For any identity arrow 1_A in $\mathcal{C}$,

$$(S f)(1_A \circ 1_A) = (S f)(1_A) = 1_{(S f)(1_A)} = (S f)(1_A) \circ (S f)(1_A)$$

□

Fact 1.2. The functor $S : \mathbf{Set} \rightarrow \mathbf{Cat}$ satisfies functoriality.

Proof. Every object set A in $\mathbf{Set}$ will get mapped to a discrete category $S A$ in $\mathbf{Cat}$. For every pairs of functions $f : A \rightarrow B, g : B \rightarrow C$ in $\mathbf{Set}$, and $a \in A, c \in C$,

$$\begin{aligned}
(S(g \circ f)) :: S a \mapsto S c &\iff g \circ f(a) = c \\
&\iff \exists b \in B : f(a) = b \wedge g(b) = c \\
&\iff \exists b \in B : S f :: S a \mapsto S b \wedge S g :: S b \mapsto S c \\
&\iff (S g \circ S f) :: S a \mapsto S c \\
S 1_A :: S a \mapsto S a &= 1_{S A}
\end{aligned} \tag{1}$$

Then functoriality is satisfied. $\square$

Proposition 1.3. The functor $S : \mathbf{Set} \rightarrow \mathbf{Cat}$ is full and faithful.

Proof. To show faithfulness, let $f, g : C \rightarrow D$ be arbitrary functions in $\mathbf{Set}$ so $S f, S g : \mathcal{C} \rightarrow \mathcal{D}$. Suppose $S f = S g$. Then since for all $a \in C$,

$$\begin{aligned}
S f :: S a \mapsto S b &\iff f :: a \mapsto b \\
S g :: S a \mapsto S b &\iff g :: a \mapsto b
\end{aligned}$$

we have

$$\forall a \in C : f :: a \mapsto b \iff S f = S g :: S a \mapsto S b \iff g :: a \mapsto b$$

and $f = g$. S is faithful.

For fullness, let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an arbitrary functor in $\mathbf{Cat}(S A, S B)$, where A, B are sets in $\mathbf{Set}$. Let $f : A \rightarrow B$. For each object $S a$ in $S A$ where $F :: S a \mapsto S b$, we can have some $a \in A, f(a) = b$ since $S A$ has $|A|$ objects. Then $S f = F$. So S is full. $\square$

Isomorphism

$$\phi : X \cong |X|$$

Define the category $S X \cong |X|$

- Objects
- Arrows
- Composition
- Unit
- Associativity Axiom
- Unit Axiom

$$S\ f : |X| \rightarrow |Y|$$

$$S\ f :: a \mapsto \phi_Y(f(\phi_X^{-1}(a)))$$

$$S\ f :: 1_a \mapsto 1_{(...)}$$

Faithful

$$\phi_Y(f(\phi_X^{-1}(a))) = \phi_Y(g(\phi_X^{-1}(a))) \implies f = g$$

Question 2

Definition 2.4. The bivariant Hom-functor is given by

$$\mathcal{C}(-, -)(A, B) := \mathcal{C}(A, B)$$

$$\mathcal{C}(-, -)(g : B \rightarrow A, f : C \rightarrow D) := h \mapsto f \circ h \circ g, \forall h \in \mathcal{C}(A, C)$$

for all objects and arrows $A, B, C, D \in \text{Ob}(\mathcal{C})$, $f, g \in \text{Ar}(\mathcal{C})$.

Proposition 2.5. *The bivariant Hom-functor is a functor.*

Proof. Let $A, B, C, D, E, F \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{C}^{\text{op}})$ be arbitrary objects.

For any object pair A, B , the functor has a mapping to the set $\mathcal{C}(A, B)$.

Let $f : A \rightarrow B, g : B \rightarrow C$ be arbitrary arrows in $\mathcal{C}$, and $h : E \rightarrow D, k : F \rightarrow E$ be arbitrary arrows in $\mathcal{C}^{\text{op}}$.

$$\begin{aligned} \mathcal{C}(-, -)(h, g) \circ \mathcal{C}(-, -)(k, f) &= (p \mapsto g \circ p \circ h) \circ (p' \mapsto f \circ p' \circ k) \\ &= (p \mapsto g \circ (f \circ p \circ k) \circ h) \\ &= (p \mapsto (g \circ f) \circ p \circ (k \circ h)) \\ &= \mathcal{C}(-, -)(h \circ k, g \circ f) \\ &= \mathcal{C}(-, -)((h, g) \circ (k, f)) \end{aligned} \tag{2}$$

So the composition law is satisfied.

For the identity laws,

$$\begin{aligned} \mathcal{C}(-, -)(1_A, 1_B) &= \mathcal{C}(1_A, 1_B) \\ &= h \mapsto 1_B \circ h \circ 1_A \\ &= h \mapsto h \\ &= 1_{\mathcal{C}(A, B)} \\ &= 1_{\mathcal{C}(-, -)(A, B)} \end{aligned} \tag{3}$$

Thus, the bivariant Hom-functor satisfies *functoriality*.

□

Question 3

(a)

Fact 3.6. Faithful functors do not, in general, reflect isomorphisms.

Example 3.7.

$$A \xrightarrow{f} B \qquad F A \xrightleftharpoons[g=(F f)^{-1}]{F f} F B$$

The diagram above shows two-object categories $\mathcal{C}, \mathcal{D}$ respectively, with the identities assumed but not drawn. The arrow $g = (F f)^{-1} : F B \rightarrow F A$ in $\mathcal{D}$ has no mapping from an arrow in $\mathcal{C}$.

Functoriality is satisfied since

$$\begin{aligned} F(1_A) &= 1_{F A}, & F(1_B) &= 1_{F B} \\ F(f \circ 1_A) &= F(f) = F(f) \circ 1_{F A} = F(f) \circ F(1_A) \\ F(1_B \circ f) &= F(f) = 1_{F B} \circ F(f) = F(1_B) \circ F(f) \end{aligned}$$

The functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is faithful since

$$F f \neq F 1_A = 1_{F A} \neq F 1_B = 1_{F B}$$

It can be noted it is not full because of the arrow g in $\mathcal{D}$.

The arrow $F f$ is isomorphic (to $(F f)^{-1}$) but f is not iso. The functor does not reflect isomorphism.

(b)

Proposition 3.8. *Every full and faithful functor reflects isomorphisms.*

Proof. Let the functor $F : \mathcal{C} \rightarrow \mathcal{D}$ be full and faithful. By fullness, for any arrow g in $\mathcal{D}$, there exists an arrow f in $\mathcal{C}$ such that $F f = g$.

Suppose we have two iso arrows g, g^{-1} in $\mathcal{C}$. Using fullness we find $f : A \rightarrow B, f' : B' \rightarrow A'$ such that $F f = g, F f' = g^{-1}$. Since g, g^{-1} are iso, $F A = F A', F B = F B'$ so

$$g : F A \rightarrow F B, \quad g^{-1} : F B \rightarrow F A$$

where

$$g^{-1} \circ g = 1_{F A}, \quad g \circ g^{-1} = 1_{F B}$$

By functoriality, we have from composition that

$$\begin{aligned} F(f \circ f') &= F f \circ F f' = g \circ g^{-1} = 1_{F B} \\ F(f' \circ f) &= F f' \circ F f = g^{-1} \circ g = 1_{F A} \end{aligned}$$

We also have from the identity law that

$$F(1_A) = 1_{F A}, \quad F(1_B) = 1_{F B}$$

By faithfulness,

$$\begin{aligned} F(1_A) = F(f' \circ f) = 1_{F A} &\implies f' \circ f = 1_A \\ F(1_B) = F(f \circ f') = 1_{F B} &\implies f \circ f' = 1_B \end{aligned}$$

So $f' = f^{-1}$, and f, f^{-1} are iso.

Given an iso arrow $F f$ in $\mathcal{D}$, we then have an iso arrow f in $\mathcal{C}$. The full and faithful functor reflects isomorphisms. □

Full and faithful functor:

$$\begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ & \longmapsto & \bullet \longrightarrow \bullet \\ \bullet & \longrightarrow & \bullet \end{array}$$

(c)

Proposition 3.9. *Every equivalence preserves monics.*

Proof. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an equivalence, and let $f : B \rightarrow C$ be a monic arrow in $\mathcal{C}$. Then $F f : F B \rightarrow F C$. Using fullness, let $F g, F h : A' \rightarrow F B$ be arbitrary arrows in $\mathcal{D}$.

By equivalence, there exists an object A of $\mathcal{C}$ such that $F(A) \cong A'$.

By fullness, let the arrows $g, h : A \rightarrow B$ be in $\mathcal{C}$.

Suppose $F f \circ F g = F f \circ F h$. Then

$$\begin{aligned} F f \circ F g = F f \circ F h &\implies F(f \circ g) = F(f \circ h) && \text{(by functoriality)} \\ &\implies f \circ g = f \circ h && \text{(by faithfulness)} \\ &\implies g = h && \text{(since } f \text{ is monic)} \\ &\implies F g = F h \end{aligned} \tag{4}$$

Then

$$\forall F g, F h : F f \circ F g = F f \circ F h \implies F g = F h$$

so $F f$ is monic and the equivalence preserves monics. □

$$\begin{aligned} F : \mathcal{C} &\rightarrow \mathcal{D}, G : \mathcal{D} \rightarrow \mathcal{C} \\ F \circ G &\cong 1_{\mathcal{D}}, \quad \eta : G \circ F \cong 1_{\mathcal{C}} \end{aligned}$$

Assume F equivalent, f monic

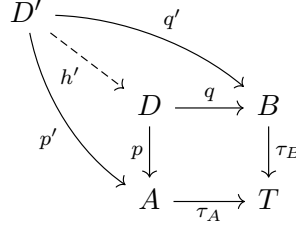
$$\begin{array}{ccc} x & \xrightarrow{\eta_x} & GF x \\ f \downarrow & & \downarrow GF f \\ y & \xrightarrow{\eta_y} & GF y \end{array}$$

$$\begin{aligned}
G(F \circ f \circ g) = G(F \circ f \circ h) &\implies GF \circ f \circ Gg = GF \circ f \circ Gh \\
&\implies (\eta_y \circ f \circ \eta_x^{-1} \circ Gg) = (\eta_y \circ f \circ \eta_x^{-1} \circ Gh) \\
&\implies Gg = Gh \\
&\implies g = h
\end{aligned} \tag{5}$$

Question 4

Proposition 4.10. *If a category $\mathcal{C}$ has a terminal object and pullbacks, it has binary products and equalisers.*

Proof. Let T be the terminal object, and let $\tau_A : A \rightarrow T$ denote the unique arrow from an object A to T .



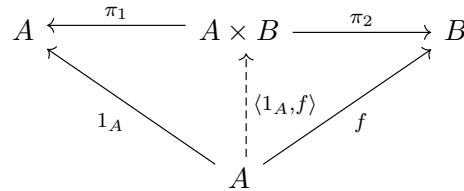
The morphisms $A \xrightarrow{\tau_A} T \xleftarrow{\tau_B} B$ then have the pullback $A \xleftarrow{p} D \xrightarrow{q} B$.

For any morphisms $A \xleftarrow{p'} D' \xrightarrow{q'} B$, we have by composition,

$$\tau_A \circ p' = \tau_{D'} = \tau_B \circ q' \implies h' : D' \rightarrow D \text{ is unique}$$

such that $p' = p \circ h'$ and $q' = q \circ h'$. Then D with morphisms $A \xleftarrow{p} D \xrightarrow{q} B$ is the product of A and B , since for all pairs $A \xleftarrow{p'} D' \xrightarrow{q'} B$, there is the unique morphism $h' = \langle p', q' \rangle : D' \rightarrow D$ such that $p' = p \circ h'$ and $q' = q \circ h'$.

Using the binary product, we can then show equalisers. Suppose we have the parallel morphisms $A \xrightarrow{f} B$. As shown before, the product $A \times B$ with pullback morphisms $A \xleftarrow{\pi_1} A \times B \xrightarrow{\pi_2} B$ exist. The following diagram with f commutes, and is similar if g is used instead of f , where 1_A is the identity on A .



Let $A \xleftarrow{p} E \xrightarrow{q} A$ be the pullback of $A \xrightarrow{\langle 1_A, f \rangle} A \times B \xleftarrow{\langle 1_A, g \rangle} A$. So $\langle 1_A, f \rangle \circ p = \langle 1_A, g \rangle \circ q$.

$$\begin{array}{ccc} E & \xrightarrow{q} & A \\ p \downarrow & & \downarrow \langle 1_A, g \rangle \\ A & \xrightarrow{\langle 1_A, f \rangle} & A \times B \end{array}$$

Then we show that $e = p = q : E \rightarrow A$ is the equaliser of (f, g) . First, we show $p = q$ using the product,

$$\begin{aligned} p &= \pi_1 \circ \langle 1_A, f \rangle \circ p \\ &= \pi_1 \circ \langle 1_A, g \rangle \circ q \\ &= q \end{aligned} \tag{6}$$

Then we have

$$\begin{aligned}
f \circ e &= \pi_2 \circ \langle 1_A, f \rangle \circ e \\
&= \pi_2 \circ \langle 1_A, g \rangle \circ e \\
&= g \circ e
\end{aligned} \tag{7}$$

Suppose we then have a morphism $h : C \rightarrow A$ such that $f \circ h = g \circ h$. Then

$$\begin{aligned}
\langle 1_A, f \rangle \circ h &= \langle 1_A \circ h, f \circ h \rangle \\
&= \langle 1_A \circ h, g \circ h \rangle \\
&= \langle 1_A, g \rangle \circ h
\end{aligned} \tag{8}$$

Then by the pullback, there is a unique morphism $\hat{h} : C \rightarrow E$ such that $h = e \circ \hat{h}$.

$$\begin{array}{ccc}
E & \xrightarrow{e} & A \rightrightarrows B \\
& \nearrow h & \uparrow \hat{h} \\
C & &
\end{array}$$

Then $e : E \rightarrow A$, the pullback of $(\langle 1_A, f \rangle, \langle 1_A, g \rangle)$, is the equaliser of (f, g) where $A \xrightarrow{f} B$.

Thus, it is shown that the category $\mathcal{C}$ has binary products and equalisers.

□

In **Set**, equalisers $e : E \rightarrow A$ have mapping to

$$\{a \in A \mid f(a) = g(a)\}$$

But this is not good enough

$$\{(a_1, a_2) \subseteq A \times A \mid f(a_1) = g(a_2)\}$$

Refine to have $a_1 = a_2$

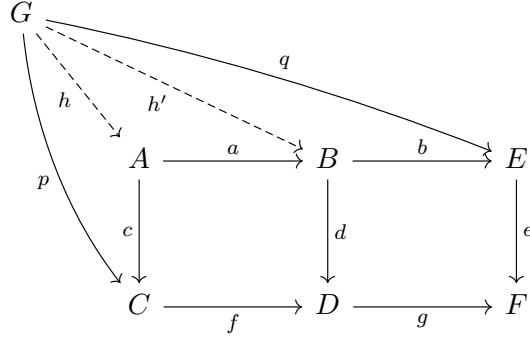
$$\{(a_1, a_2) \subseteq A \times A \mid a_1 \wedge f(a_1) = g(a_2)\} \cong \{a \in A \mid f(a) = g(a)\}$$

Also equaliser

$$\begin{array}{ccc}
E & \longrightarrow & B \\
\downarrow & \langle 1_B, 1_B \rangle \downarrow & \\
A & \xrightarrow{\langle f, g \rangle} & B \times B
\end{array}$$

Question 5

Proposition 5.11. *If $ABCD, BEDF$ are pullback squares, then so is $AECF$.*



Proof. Suppose we have morphisms $C \xleftarrow{p} G \xrightarrow{q} E$ such that $(g \circ f) \circ p = e \circ q$.

By the pullback square $BEDF$,

$$e \circ q = g \circ (f \circ p) \implies \exists! h' : G \rightarrow B, q = b \circ h' \wedge f \circ p = d \circ h'$$

By the pullback square $ABCD$ and using h' ,

$$d \circ h' = f \circ p \implies \exists! h : G \rightarrow A, p = c \circ h \wedge h' = a \circ h$$

From the unique h, h' such that $q = b \circ h', h' = a \circ h$, then h is the unique morphism where $q = b \circ a \circ h$. Then combining $q = (b \circ a) \circ h$ with $p = c \circ h$ (from pullback square $ABCD$) where h is unique, $AECF$ is also a pullback square. $\square$

Question 6

The diagram D maps as follows:

$$D :: n \mapsto ([n], \leq), \quad (f : n \rightarrow m) \mapsto (g : ([n], \leq) \hookrightarrow ([m], \leq))$$

where g is the inclusion map from $[n]$ to $[m]$.

$$\mathcal{J} \xrightarrow{D} \mathbf{Natural}$$

$$\begin{array}{ccc} n & \xrightarrow{\quad} & ([n], \leq) \\ f \downarrow & \xrightarrow{\quad} & \downarrow g \\ m & \xrightarrow{\quad} & ([m], \leq) \end{array}$$

Each cone to D is an object $([n], \leq)$ in $\mathbf{Natural}$ and the family of arrows γ ,

$$\{\gamma_I : ([n], \leq) \rightarrow (D I)\}_{I \in \text{Ob}(\mathcal{J})} = \{\gamma_m : ([n], \leq) \hookrightarrow ([m], \leq)\}_{m \in \mathbb{N}, m \geq n}$$

Let $([p], \leq)$ with its family of arrows δ be an arbitrary cone to D . Since inclusion maps from $[p]$ to $[q]$ are only defined when $p \leq q$, consider $f : n \rightarrow m$ in $\mathcal{J}$, where $p \leq n \leq m$. We can see that

$$\forall i \in [p] : (D f) \circ \delta_n(i) = (D f)(i) = \delta_m(i) = i$$

so $(D f) \circ \delta_n = \delta_m$.

$$\begin{array}{ccc}
 ([n], \leq) & \xrightarrow{D f} & ([m], \leq) \\
 & \nwarrow \delta_n \quad \nearrow \delta_m & \\
 & ([p], \leq) &
 \end{array}$$

The limit for D is the object $(\mathbb{N}, \leq)$ with its family of arrows containing only the identity, $\{1_{\mathbb{N}}\}$. For any other posets $([n], \leq)$, $n \in \mathbb{N}$ in **Natural**, $[n] \subset \mathbb{N}$, so there is no mapping from $(\mathbb{N}, \leq)$ other than to itself.

$$\begin{array}{ccc}
 & (\mathbb{N}, \leq) & \\
 \gamma_{\mathbb{N}} \nearrow & & \nwarrow 1_{\mathbb{N}} \\
 ([n], \leq) & \xrightarrow{\gamma_{\mathbb{N}}} & (\mathbb{N}, \leq)
 \end{array}$$

The ~~ee~~limit for D is the object $([0], \leq)$ with its family of arrows

$$\{\gamma_n : ([0], \leq) \hookrightarrow ([n], \leq)\}_{n \in \mathbb{N}}$$

For any other posets $([n], \leq)$, $n \in \mathbb{N}^+$ in **Natural**, $[0] \subset [n]$, so there is no mapping to $([0], \leq)$ other than from itself.

$$\begin{array}{ccc}
 \Delta_x i & \xrightarrow{\pi_x} & J i \\
 \Delta_x f \downarrow & & \downarrow J f \\
 \Delta_x j & \xrightarrow{\pi_j} & J j
 \end{array}$$

$$\begin{array}{ccc}
 & W & \\
 & \downarrow & \\
 & \lim_I J & \\
 \swarrow & & \searrow \\
 J i & \xrightarrow{J f} & J j
 \end{array}$$

Using $\Rightarrow$ as natural transformation

$$\Delta_X \Rightarrow J$$

$$W \rightarrow X$$

$$\Delta_W \Rightarrow J$$

There are many maps to 0 which makes it not commute.

There is a unique map from 0 to each set.

$$(\mathbb{N}, \leq) \longrightarrow \mathbf{Pos}$$

$$\begin{array}{ccc} n & & [n] \\ \downarrow \leq & & \downarrow \\ m & & [m] \end{array}$$